



ABES Engineering College, Ghaziabad
Department of Computer Science and Engineering (CSE2)

Session: 2026-27 Semester: ___III___ Section: __CSE-21-CSE26___

Subject Code: 25CS303

Subject Name: Advanced Data Structure

Assignment 1

Date of Assignment: 27/08/2026

Date of submission: 03/08/2026

S.No.	KL, CO	Question	Marks
1	K2, CO1	Define a stack and explain its LIFO property with a real-life example. Describe array-based and linked-list-based representations, with advantages/limitations of each.	5
2	K3, CO1	Write algorithms for push, pop, and peek operations on an array-based stack, handling overflow/underflow conditions.	5
3	K3, CO1	a) Convert $(A + B) * (C - D) / E$ from infix to postfix, showing stack contents at each step. b) Convert $A + B * (C - D) / E$ from infix to prefix, showing intermediate steps.	5
4	K3, CO1	Define a queue and its FIFO property. Differentiate linear queue vs. circular queue and justify why circular queues are preferred. Write algorithms for enqueue and dequeue operations on a circular queue, correctly detecting full/empty conditions.	5
5	K3, CO1	Explain Stack and Queue implementation in C++ STL; list member functions and demonstrate with a program.	5
6	K5, CO1	Customers arrive and are served on a first-come-first-served basis, but VIP members (with pre-paid membership) must always be served before regular customers, regardless of arrival time. Among VIP members, earlier arrivals are served first; the same applies among regular customers. (i) Identify which data structure(s) from the syllabus (queue, circular queue, priority queue, dequeue) would best model this system. Justify your choice. (ii) Design the structure of a node/record to store each customer's data along with the necessary priority information.	5